



Exploring relationships between nation branding and foreign direct investment

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Revised: 9 March 2022 / Accepted: 10 March 2022 / Published online: 26 March 2022
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Abstract

This study explores the influence of nation branding on a country's ability to send and receive foreign direct investment (FDI). Employing Anholt's Nation Brand Index (NBI) and FDI data, the relationship is analyzed using valued exponential random graph models (ERGM) to account for interdependence between FDI flows while simultaneously estimating the changes in FDI flows in respect to differences in NBI scores. The study finds that while a stronger nation brand does contribute positively to both incoming and outgoing FDI flows, there are important differences in how different categories of nation branding contribute to FDI flows. Namely, the Export category is strongly associated with outgoing FDI, the Investment, Government, and Culture categories are positively associated with incoming FDI, and that the People and Tourism categories are negatively associated with incoming FDI. This study contributes to the nascent literature on place branding in that it combines the prominent Eclectic Paradigm for FDI from the economics literature with marketing and communication literature's theories on nation branding.

Keywords Nation branding · Foreign direct investment · Network analysis · Country-of-origin effect

Introduction

In his book *Brand New Justice*, Simon Anholt (2006) proposed that nation branding can be an important element for economic development and greater prosperity. Anholt argues that countries can benefit from developing a strategic national brand strategy that is communicated consistently and broadly around the world because it allows countries to export quality branded products to consumers in richer nations and also attract inward investment from other countries. According to Anholt, if done well, nation branding can make a difference in parts of the world that really need help.

Anholt's premise has been met with considerable skepticism. The idea of selling a country with a catchy tagline or paying advertising consultants millions of dollars to create TV spots promoting a nation may not be the best approach for governments to help their people (Aronczyk 2013). The scholarly debate has been fierce for several decades, creating

two camps in the international public relations and public diplomacy literature—one that supports the idea of nation branding and another that eschews the notion (Fullerton and Kendrick 2017). Anholt and others have provided well-documented case studies and personal anecdotes to make their argument, but there are only a few large-scale empirical studies that address the question—can nation branding influence a country's economic well-being?

This study explores this question quantitatively using social network modeling to measure the relationship between aspects of nation branding and outgoing and incoming foreign direct investment (FDI). FDI has the potential to contribute to the development and general economic performance of a country and nearly every country actively seeks to attract it Narula and Driffield (2012). The analysis uses the Anholt-Ipsos Nation Brand Index (NBI) data for a nation's brand—a consumer opinion survey that captures perception of foreign countries on a six-dimensional scale¹—and bilateral FDI data as the dependent variable. While other nation brand indices exist (see for example Futurebrand Country Brand Index²), Anholt-Ipsos (formerly Anholt-GfK Roper Nations Brand Index) is among the oldest, largest

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¹ <https://www.ipsos.com/sites/default/files/anholt-ipsos-nation-brands-index.pdf>.

² <https://www.futurebrand.com/futurebrand-country-index>.



(both in terms of sample size and frame), and most well-known. Given that Anholt established the original concept of nation branding and the nation brand hexagon, the Anholt-Ipsos index, based on his seminal work, provides the most applicable data for this study. Other similar studies involving FDI and nation branding (e.g. Lahrech et al. 2020; Kalamova and Konrad 2010) have also used the Anholt data set.

Literature review

Place branding can be defined as the “practice of applying brand strategy and other marketing techniques and disciplines to the economic, social, political and cultural development of cities, regions and countries” (Kavaratzis and Ashworth 2006, p. 183). Essentially, nation branding (or country branding) can be understood as applying the marketing concept of branding to nations. Fullerton and Kendrick (2017), in their Model of Country Concept, posit that nation branding is a management and communication function that attempts to shape attitudes about Country A in the minds of citizens in Country B.

Anholt (2004) originally put forth the idea of nation branding using his “Place Branding Hexagon”. The six points of the hexagon represent ways in which countries have contact with people in other nations—Tourism, Brand Exports, People, Culture, Governance and Investment/Immigration. It is in the Investment/Immigration part of the hexagon that FDI resides. Therefore, the nation-branding hexagon can be a helpful tool to understand the relationship between a country’s nation brand and FDI.

FDI is a financial investment made by a firm or individual in one country into business in another country. An example of FDI might include Nissan, a Japanese car company, building an automobile manufacturing plant in Mexico or McDonald’s opening a restaurant in Russia. According to the economics literature, FDI creates new markets, more customers, and lower costs for the firm that is investing. It also has the potential to benefit the host country by stimulating economic development, providing employment opportunities, transferring technology, and training a local workforce (Kurtishi-Kastrati 2013). Therefore, developing countries are typically eager to receive FDI from more developed countries and they compete, to the extent that it ‘borders on the obsessive’, to garner these investments (Narula and Driffield 2012, p. 1).

NBI and FDI

Jacobsen (2012) proposed a model to understand the relationship between place branding and inward investment. Using the product brand equity model (Keller 1993) as a basis, Jacobsen recognized that places could be marketed

like products and branding principles could be applied to places. Therefore, place brands, like product brands, may possess tangible (quality, impression, promotion) and intangible (awareness, heritage, image, personality, reputation and confidence) attributes as well as functional (lifestyle, education, real estate market) and symbolic (prestige and distinction) place brand benefits. According to the Jacobsen model, these attributes and benefits combine to influence investment consideration and investment behavior. Bose et al. (2016) continued the analogy of product brands to place brands by connecting the well-established concept of customer-based brand equity (CBBE) to places and establishing a measurement tool for a new variable—customer-based place brand equity (CBPBE). The measurement tool allowed Bose and others (Bose et al. 2021) to later test the impact of customer-based place brand equity on tourism destination loyalty and investment attractiveness. These studies used survey data from consumers and investors in their analysis.

The relationship between NBI and FDI using linear regression models has been examined in at least two previous studies (see Lahrech et al. 2020; Kalamova and Konrad 2010). In both of these studies, the relationship was generally found to be positive. However, these previous studies had a more limited focus than the current paper expands. Lahrech and colleagues (2020) only included the countries with the ten highest NBI scores. This approach, while providing insight into what a nation brand means for top branded countries, both limits the statistical power of the investigation and generalizability of the findings. Second, Lahrech et al. used topline NBI scores and did not investigate differences between the different nation brand index categories (i.e. Tourism, Government). This paper includes all available countries and breaks down each category of NBI’s impact on FDI flows. Kalamova and Konrad (2010) included more countries in their sample and broke down NBI by category in their additional models as robustness checks, but their study only estimated NBI’s relationship with incoming FDI and relied on linear models that fail to account for interdependence between the observations.

FDI promotion

Most countries have an Investment Promotion Agency (IPA) as part of its strategy to attract FDI. Some studies suggest that the promotional efforts of IPAs, especially promotional materials and public relations tactics such as business forums and international meetings, tend to work such that every dollar, above a certain threshold, that a country spends on FDI promotion yields returns, indeed better returns than subsidy or tax credits (Morisset and Andrews-Johnson 2004). However, other studies



of specific FDI promotional tactics have produced mixed results in generating FDI. Ayuningtias (2018) tested Indonesia's "Invest in Remarkable Indonesia" television commercial among a small sample of New York investors and found that while the commercial did not shift attitudes toward investing in Indonesia in pre/post viewing, a majority of investors said after viewing the commercial that the spot created awareness that Indonesia was "a welcoming and attractive place to invest" (Ayuningtias 2018, p. 61). Kindra et al. (1998) reported survey results that promotion activities such as advertising did not influence FDI generation in the ASEAN region, but the host country having a pro-business reputation or image did. To the extent that targeted promotional activities featuring a country as a good place to invest contributes to a country's nation brand, having a positive and well-communicated nation brand may be one way to make a country more likely to attract FDI.

Theoretical framework

The Country-of-Origin effect (COE) is a widely studied theory in the field of business going back to the 1960s (Schooler 1965). COE describes how consumers use a product's "made in" label to assess product quality and make purchase decisions. A country's image, which refers to consumers' perceptions of a specific country and is an element of the nation's brand (Fullerton and Kendrick 2017), reflect COE (Mariutti 2017). The image that consumers hold in their minds about countries can be considered a product feature and transferred to the product's quality as a sort of heuristic or mental shortcut in the decision-making process (Li and Wyer 1994). COE is found to be "somewhat generalizable" (Peterson and Jolibert 1995, p. 895) and a "substantial factor" (Steenkamp and Verlegh 1999, p. 538) in product evaluation, perceived quality, attitudes, and purchasing intention. There are many common examples of COE, such as Swiss chocolate and French wine being considered of the highest quality by consumers around the world. Liu et al. (2017) found that Chinese consumers' wine preference was driven by their COE perceptions of Australia and France. Product knowledge and ethnocentrism attenuated the effect. In an extensive review of the literature, Mariutti (2017) explored the evolution of COE to current research regarding nation branding and country reputation and found evidence for conceptual connections.

The Eclectic Paradigm of FDI is one of the most used theoretical frameworks for why firms send and receive FDI. It states that firms choose to send FDI because of ownership, location, and internationalization advantages (Dunning 1988). The location advantage largely refers to FDI overcoming economic barriers to trade, such as transportation costs, or policy barriers, such as tariffs. Internalization is

more complex, stating there are complex reasons a firm cannot simply license or provide technology to a local firm to produce their goods.

The ownership advantage, which is most relevant to this study, is defined as "spatially transferable intangible assets of the parent companies" (Dunning 2001, p. 174). This can include aspects such as production processes, but it can also refer to a firms' branding. COE explains how the firms' brand can be influenced by the nation brand. Taken together, COE and the Eclectic Paradigm provide a theoretical framework to demonstrate how NBI and FDI may be related. Firms leveraging their ownership advantages from a national brand identity will then seek location cost advantages through FDI to reduce costs while still maintaining final control over final products. For example, Japanese vehicles are world famous for their quality. However, after their brand was established and wages began to rise in Japan, they relied on offshoring labor intensive production of intermediate goods to maintain a comparative advantage, as many industrialized countries have done (Baldwin 2006). This combination of strong branding and cost advantages allowed their share of global car production to continue to rise, despite a greater percentage of the vehicle being made outside of Japan. This practice is possible and does not harm the brand's image, in part, because end-consumers have little knowledge of global supply chains for intermediate goods.

This pattern between branding and FDI matches the U-shaped pattern between financial development and exports (Zhao et al. 2017). As a country's financial development and brand increases, so do their exports. As exports rise and the economy develops, there is a need to reduce costs to maintain comparative advantage. Therefore, firms turn to FDI to offshore production, reducing net exports, all the while continuing to leverage the developed brand strength for the final product. This makes sense, as only large firms, with the most established brands, can engage in high fixed cost FDI driven offshoring, much like and complementary to the New-New Trade Theory that explains why most of international trade is also driven by the largest and most productive firms (Melitz 2003). cursory examination of FDI patterns bears this out. In the last year of data in this study, 2007, over 60% of FDI came from only 10 countries (International Monetary Fund 2021).

Therefore, the first hypothesis for this study is that *countries with a higher NBI index score for the Export category will send more FDI*.

When deciding where to send FDI, firms initially keep production close to home for greater control, familiarity, and lower transportation costs. For example, US firms offshored to places like Mexico and Japanese firms moved production to SE Asia (Baldwin 2006). However economic considerations such as transportation costs are not the only things firms consider. While outright expropriation is rare,



firms do consider general risks of doing business in a country and weigh these against economic advantages (Graham et al. 2018). This was seen recently with gradual decoupling of Western markets with an increasingly competitive China (Economist 2021). While traditional measures of risk such as institutional strength and polity are useful, and included in this model, they do not completely capture risk nor do they have a clear relationship with firm preferences. In fact, recent literature has found that firms are adept at overcoming risk through strategies such as local partnerships and may even prefer traditionally riskier markets due to greater chances for preferential treatment (Johns and Wellhausen 2016). Therefore, a potential way to measure risk perception beyond traditional measures is the branding of a country. Particularly, branding regarding government, investment, and culture would capture risk perception of a country.

Therefore, the second hypothesis is that *countries with a higher NBI index score for the Investment, Culture, and Government categories are more likely to receive FDI*.

Data and research design

FDI data

The bilateral FDI data from UNCTAD (2014) is used in this study as the dependent variable for several reasons. First, it allows use of social network models that control for structural dependence between observations. Ignoring structural dependence in FDI networks has been shown to lead to biased estimate (Schoeneman et al. 2020). Second, the use of bilateral data allows the study to leverage the network structure to estimate the impact of nation branding scores on both the inflow and outflow of FDI, as well as how levels of nation branding scores change structural effects.

Although interested in changes in FDI, the researchers elected to use FDI stock as the dependent variable. This is because the model used to estimate FDI cannot accommodate negative dependent variable values and so when transforming negative values to zero, changes are minimized since negative FDI stock values only occur when debt to a parent company exceeds total stock value. To still effectively model FDI flows, a lagged dependent variable is included in the analysis. The study also logs the FDI stock value and multiplies it by two before rounding to whole number, so that the data can be used in a count model. This reduces the influence of outliers on coefficient estimates while maintaining a sufficient amount of variation. As demonstrated in past work (Schoeneman et al. 2020), this transformation does not significantly change the correlation between FDI and other variables

Nation branding index variables

The nation branding variable used in the model comes from Anholt's Nation Brand Index (Anholt 2005) and was provided directly by Simon Anholt. These data are measured quarterly from 2005 to 2007 and include an aggregate nation brand index score as well as a disaggregated score following Anholt's hexagon categories. The scores are based on rank-ordered responses about countries from online consumer surveys administered to stratified random samples within each country measured. Samples are stratified by age, gender, and where applicable, geographical region, race and ethnicity. Sample size ranged from 200 to 1000, with a mean of 740 and median of 1000. Respondents do not rank their home country. Several transformations to the data were made before including it in the model. First, to control for different scales of the surveys when given, each quarterly survey scores were normalized between one and zero. Second, to match the data with yearly FDI data, scores for all quarters a country was included were averaged for each year.

The six categories that a nation brand is scored and disaggregated by are Exports, Government, Investment, Culture, People, and Tourism. The questions used for each category were designed to capture the essence of the category and not to record general rankings. Export questions were attempting to measure satisfaction and country-level association with said country's exported goods. For example, respondents were asked to 'Please pick the category of product or service which you would most expect to be produced in each country'. Government attempts to measure how competently and fairly a country is governed. Investment measures the perceived economic and social conditions of a country. Culture measures the popularity of a country's cultural heritage as well as a country's ability to produce winning sports teams. The People category measures how people in a country are perceived to be welcoming and well-liked. And lastly, and most straight forward, Tourism measures how much people want to visit a country for leisure travel purposes.

Exogenous control variables

In addition to the NBI categories, additional economic and political control variables are included. These include 'gravity' variables, which are the log product of the dyad's real GDP and logged Euclidean distance. GDP is expected to positively correlate with FDI due to a larger market and distance is expected to be negatively correlated with FDI due to increased costs (Kahouli and Maktouf 2015). For the purpose of model convergence, the logged product of dyadic GDP has been estimated as one variable in the model, rather than being estimated separately. Additional nodal economic controls include origin and destination countries' trade openness (trade as % of GDP) and GDP per capita as a control



for a country's level of dependence on global markets and level of development. For edge level economic controls, the model includes bilateral trade volumes to control for the complementary relationship between FDI and trade (Aizenman and Noy 2006). GDP and population data are from the *Penn World Table* (Feenstra et al. 2015), trade openness data are from the World Bank's *World Development Indicators*, and trade volume data are from the OECD (2016).

The literature identifies a number of institutional variables at the nodal and edge level that are also included in the model as controls. At the nodal level, origin and destination polity is included (Marshall and Jagers 2010). While inconclusive, there is large body of literature that investigates whether democratic institutions are positively associated with FDI inflows (see, e.g., Jensen 2003; Jakobsen and De Soysa 2006; Li et al. 2018; Wright and Zhu 2018; Arel-Bundock 2017; Busse and Hefeker 2007). At the edge level the model includes alliance and defense treaties, trade and investment agreements, and a homophily variable for OECD membership. Military treaty variables, from Gibler (2009), are expected to be positively associated with FDI flows due to amicable political relations that reduce expropriation risk (Li and Vashchilko 2010). Bilateral investment treaties (BITs), are included as a binary variable coded as one if the pair has an agreement that includes investment policy, which are expected to increase FDI flows by removing barriers (e.g., Büthe and Milner 2008; Allee and Peinhardt 2011; Medvedev 2012; Osnago et al. 2016). For trade agreements, which have been shown to be positively associated with FDI (Büthe and Milner 2008, 2014), the model includes a preferential trade agreement depth variable that utilizes latent trait analysis with 48 binary input variables regarding what the agreement covers (Dür et al. 2014).

Network model and variables

Modeling choice and specification roughly follows that of Schoeneman et al. (2020), which demonstrates that using the count-valued exponential random graph model (ERGM) (Krivitsky 2012) is robust for estimating and controlling for structural dependencies in FDI networks. In addition to controlling for dependency, this model offers several advantages over other options currently available. First, the model is capable of controlling for zero-inflation, which is large in FDI networks. Second, the structural network coefficients are estimated at the network level, allowing comparison of magnitude of these effects between different networks. A third reason for using ERGMs is that they have been growing in popularity in the communication literature and thus scholars in the field are becoming more familiar with them (Pilny and Atouba 2018; Song 2015). One thing to note is that just like the structural dependencies, all of the aforementioned variables are modeled at the network level and

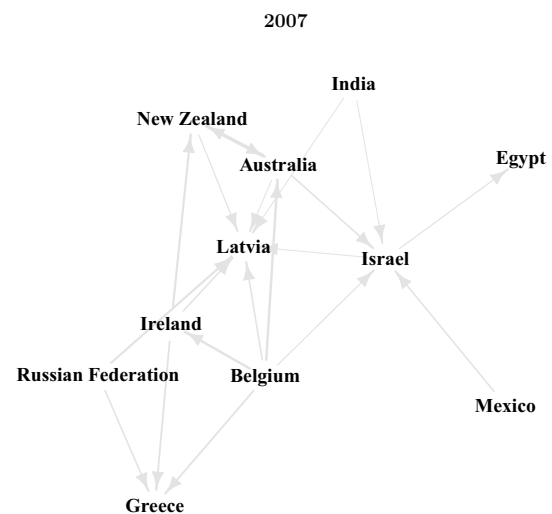


Fig. 1 Network Plot of FDI Stock. Arrows represent the direction of FDI and line thickness represents volume

incorporated into the ERGM probability distribution (Desmarais and Cranmer 2012).

For network effects, count-valued ERGMs do not have as many terms available as binary ERGMs. However, researchers are able to include a number of statistics that account for the shape of edge distributions as well as two key structural dependencies. For the shape of the edge distribution, this model includes the sum of edge values, which is analogous to an intercept, the sum of square-root values of edges to model dispersion, and the number of non-zero edges to model the frequency of zero-valued edges. The first structural dependency statistic included is reciprocity, or the likelihood that when a country i increases the FDI it sends to country j , country j will increase the amount of FDI it sends to country i . The second statistic included is transitivity, also known as the clustering coefficient. Transitivity measures the likelihood that country i sends more FDI to country j when there the directed two-paths going from country i to country k to country j increase in value. While there are several options for specifying this, in this study researchers chose that *min-max-min* specification that uses the minimum value of i to j for the largest minimum value along two-paths between i and j , as it is the most stable and conservative.

Figure 1 shows a subset of the FDI network to illustrate the structures modeled. The plot illustrates the FDI network for non-isolates after subsetting the 2007 network to nodes that have 40 or fewer ties. Researchers subsetted this way because it reveals local networks more clearly, but still includes the structures that are representative of the network at large. The first feature seen in the plot is that for the strongest tie—the tie between New Zealand and Australia—the tie is reciprocal. This is a consistent feature of the FDI networks. The second feature that is seen is triadic closure





Fig. 2 Estimates of terms in Poisson ERGMs

when countries are more closely clustered in the network. An example of this is the two-path from Belgium to Greece through Ireland coupled with the direct path from Belgium to Greece. Lastly, while some cyclical triads may form due to transitive driven ties going in multiple directions, in the subset this is not the dominant triad classification even for a subset that has fewer overall ties. The few cycles that do occur from overlapping transitive ties are accounted for through the transitive weights term.

Two different model specifications are estimated. First, the researchers ran separate models for each year with all control variables and network terms using the aggregated form of the NBI scores. Second, researchers estimate a count ERGM using all six NBI category scores along with all other control variables. For this model, a time-pooled ERGM is used for greater statistical power and to average out fluctuations due to time and sample.

Results

As seen in Fig. 2, the ERGM results for overall FDI agree with previous studies (Lahrech et al. 2020; Kalamova and Konrad 2010) that nation brand is positively correlated with FDI inflows.³ However, the results of this study show that the coefficient for NBI index for sending FDI is substantially larger than receiving. Previous studies that did not account for both outflows and inflows, and thus did not account for interdependence between the two, misses this feature. To understand why the effect size is so different, it is necessary to look at the disaggregated coefficients in Fig. 3. What is seen in the breakdown is that coefficients for the origin of FDI are either close to zero or largely positive, allowing the overall coefficient to be large. However, the effects of different categories for receiving FDI are more evenly split on each side of zero, which means the net effect would be closer to zero. These differences are hugely important, as it demonstrates to policymakers that simply investing in overall branding is not the best path to increasing FDI flows.

Rather, focusing on specific brand categories will have a greater return on investment. Furthermore, increasing some categories could potentially have a negative impact on their ability to attract and send FDI or at least identify the causes of why certain places are not perceived as attractive investment destinations.

When breaking down NBI category coefficients for sending FDI, a country's brand for Exports is the largest positive predictor for sending FDI. This supports the first hypothesis that firms in countries with established brands can leverage their brand image, or COE, to continue to sell at competitive prices by sending FDI to establish more efficient supply chains. Following Export Brand, it can be seen that Tourism and People are significant and positive, indicating that countries with strong branding in these categories are well-known and more likely to be able to form strong relationships with other countries to receive their firms' FDI. The remaining three categories, Investment, Government, and Culture, are close to zero, indicating they are less important for firms' ability to send FDI.

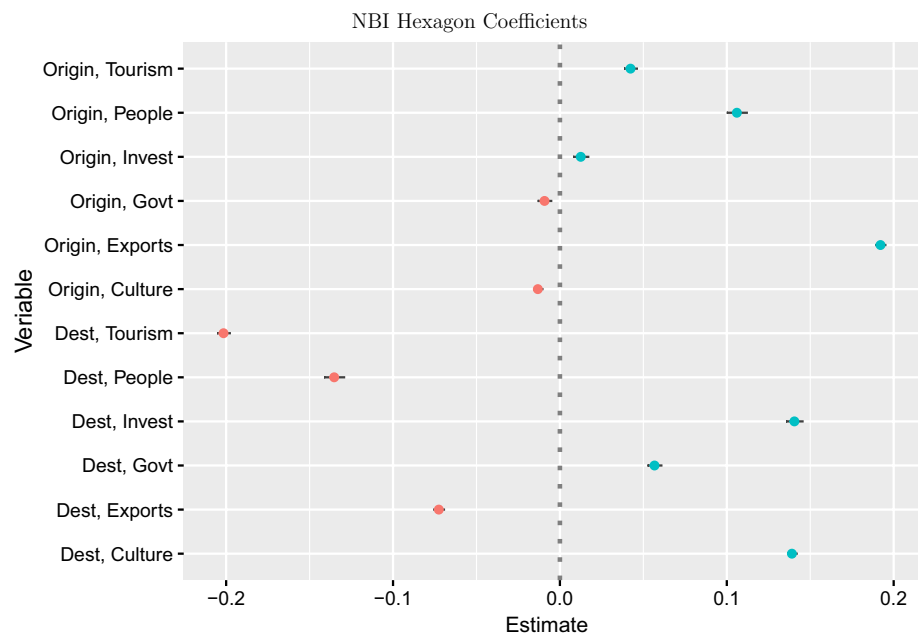
For destinations of FDI, the strongest positive correlations are for Investment and Culture, followed by Government. This is consistent with hypothesis two, that investors look for countries that are perceived to be safe from expropriation risks, as strong nation brands for Investment, Government, and Culture scores would be correlated with low-risk perceptions. These results also mean that even after controlling for measures of policy such as BITs, PTAs, and polity, consumer perceptions about the country still drive investment decisions, even if they have the same theoretical motivation as the policy variables. Another interesting insight is that coefficient for Culture is roughly double that of Government. This demonstrates that country image can still impact FDI decisions beyond that of government-based institutional protection. This finding further supports the concept established in past literature that promotion of a country brand, when it relates to specific categories, can influence firms' decisions about whether or not a country is a good place to invest.

In the ERGM results of this study, the strongest negative predictor for receiving FDI is Tourism, which is contrary to other studies that do not use NBI data. An explanation for this finding is that popular tourist destinations are generally less likely to have established manufacturing industries for offshoring. The next strongest negative predictor is People, followed by Exports. This seems counter intuitive, but this is likely not because these countries have unlikeable people or products, but because they are less well-known (i.e. a weaker COE). Places that are less-well known and also not popular tourist destinations are also more likely to have lower costs for inputs such as labor and land, making offshoring more attractive. Furthermore, export reputation in global supply chains would follow a

³ Full model results are provided in the Appendix.



Fig. 3 Estimates of terms in Poisson time-pooled ERGMs. Bars span 95% confidence intervals. Bars span 95% confidence intervals. For some models, the confidence intervals are not visible due to being small and the large range of the coefficient estimates



hierarchy. Countries with the stronger brand send FDI, while those with weaker brands produce intermediate goods as their brand image only matters to industry insiders. Because of the nature of the sample, which is broadly consumers that only focus on the final product, intermediate export brand among industry insiders might be positive for FDI, but would not be reflected in the results. The production location for many inputs for a final product produced via FDI will never be perceived by final consumers.

This finding for the relationship between NBI and tourism differs from some past work that assumes a positive relationship between place branding measurements for tourism and investment (Jacobsen 2012; Bose et al. 2021), although it is consistent with the previously mentioned article from Kalamova and Konrad (2010). There are two main explanations for this difference. First, the sampling frame is quite different. Anholt's survey is conducted with customers across 40 countries with a median sample size of 1,000. The works cited have samples from one or two countries, survey investors instead of consumers, and sample sizes around 200. Second, the dependent variable is recorded FDI stock instead of using an attitudinal scale for of how important a variable is for investment, according to surveys of investors. While investors will accurately identity important issues for investment decisions, this could differ from actual volume of FDI flows. FDI volume follows patterns from the aforementioned structural dependencies, which are unlikely to be identified by investors. Furthermore, FDI volume, like much of economic activity, follows a power-law distribution as only the most productive firms engage in foreign activities (Helpman et al. 2004) and using surveys of investors to construct the

dependent variable would give every survey respondent equal weight. This finding and explanation is consistent with Bose's suggestion that "investors may consider specific attributes of regional identity that do not necessarily relate to tourism for choosing the 'place' as the investment destination" (Bose et al. 2021, p. 11).

Combined, these results lead to a more general, but nonetheless interesting insight about the interactions between the Eclectic Paradigm (Dunning 2001) and nation branding and COE. The robust correlations between a number of NBI categories for both outgoing and incoming FDI flows indicate that the same way consumer-based brand equity drives consumer purchasing decisions, place brand perceptions drive investment decisions. The NBI is built on consumer perceptions of a country and its people, not differences in economic conditions. If we think of investors as consumers, they are using their perception of a country's NBI categories as informational shortcut by which to determine the quality of an investment destination. This has important implications for future research and policymakers as it works toward a more developed theoretical framework for understanding FDI flows.

Conclusion

Through the use of frontier social network analysis methodology and the marriage of the popular Eclectic Paradigm for FDI from the economics literature with nation branding theories including COE and consumer-based place brand equity, this work makes a number of important and novel contributions to the literature of nation branding and FDI



while engaging seminal extant literature. For example, while the findings are mostly consistent with other studies that examine the relationship between FDI and NBI, this study builds on and extends others' work by using novel network modeling to account for interdependence while simultaneously measuring NBI's relationship with both outgoing and incoming FDI. Doing so reveals important differences in the results. Namely that while overall NBI does contribute to increases in FDI flows, there are significant differences in magnitude of its effect on outgoing and incoming FDI. Additionally, when NBI is broken down by category, the models show not only differences in magnitude, but differences in directions across categories.

To elucidate how important it is to model the network in a way that accounts for these categorical differences, one can look at the subset of the 2007 network presented in Fig. 1. Latvia, which has the second lowest overall NBI score, scores relatively higher for its investment brand score, which explains why it more central as recipient of FDI in the subset of the network. Furthermore, Latvia has the lowest export score of the subset, and consequently does not send any FDI within the subset. Other examples include Greece's relatively higher score for investment than exports, which explains why it receives FDI from three countries in the subset but does not send FDI to any of countries within the subset and Belgium's relatively higher score for exports than investment, which explains why it sends so much more FDI within the sample than it receives.

Another important attribute of the modeling approach is that it controls for interdependence within the network and can handle large, global samples to model the entirety of the network, controlling for global interdependence and producing generalizeable results. Taken together, these findings motivate further study to include the use of network modeling, as it demonstrates the need to understand foreign direct investment's networks structural dependencies.

Overall, the results show that there is a strong and positive relationship between a country's nation brand and foreign direct investment, even after controlling for logical variables such as GDP, location and political ties between countries. This supports Anholt's assertion that nation branding is good for countries, albeit not all points on the Nation Brand Hexagon contribute equally to increasing a country's FDI flows. According to the findings from this study, it is countries with a strong nation brand for Exports that are best able to offshore parts of their manufacturing, via FDI, to other countries where production is more affordable due to the products' brand among end consumers. At the same time, countries with strong nation brand for Government, Investment and Culture are able to attract FDI, which can directly lead to stronger economies and better jobs for the people of that country. These findings confirm the vast body of research on COE, which deals with the perception of quality for a nation's product exports.

There are several limitations to this study that future researchers should seek to address. First, the data were collected more than a decade ago. The time-frame 2005-2007 provided a good period of analysis—it was well after the economic disruptions of the 9/11 attacks and prior to the global economic downturn of 2008. Future studies should analyze similar current data to confirm these results when it becomes available. Second, future studies should expand the sample of countries included. NBI data are currently only collected for large markets, which primarily includes industrialized and rapidly industrializing countries. While these countries account for most of global FDI flows, this limits studies understanding of how lesser developed countries can best attract FDI - an important theme that Anholt addressed in his 2005 book.

Summary statistics

See Fig. 4.



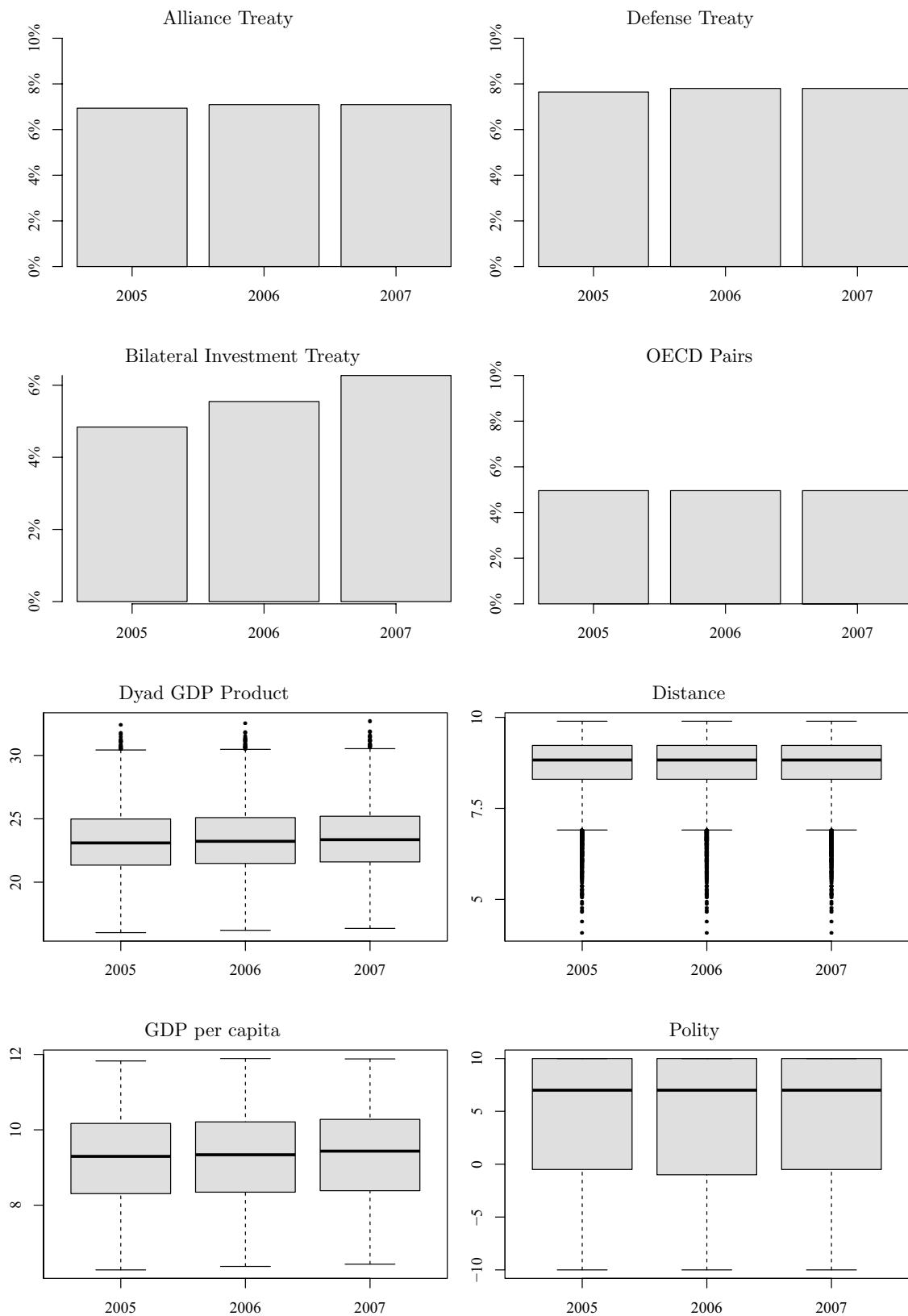


Fig. 4 Frequency count bar-plots for binary variables and distribution plots for continuous variables



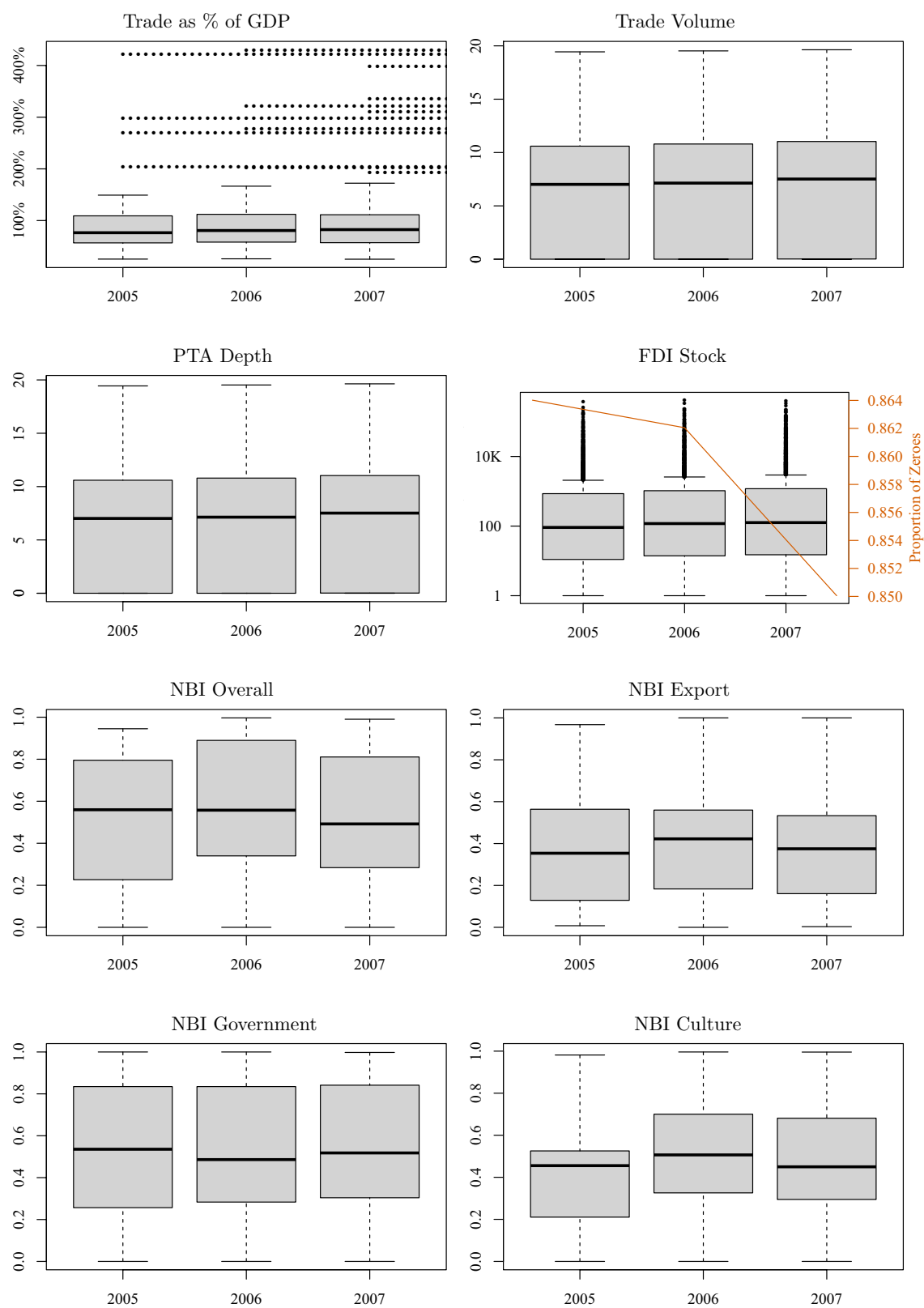


Fig. 4 (continued)



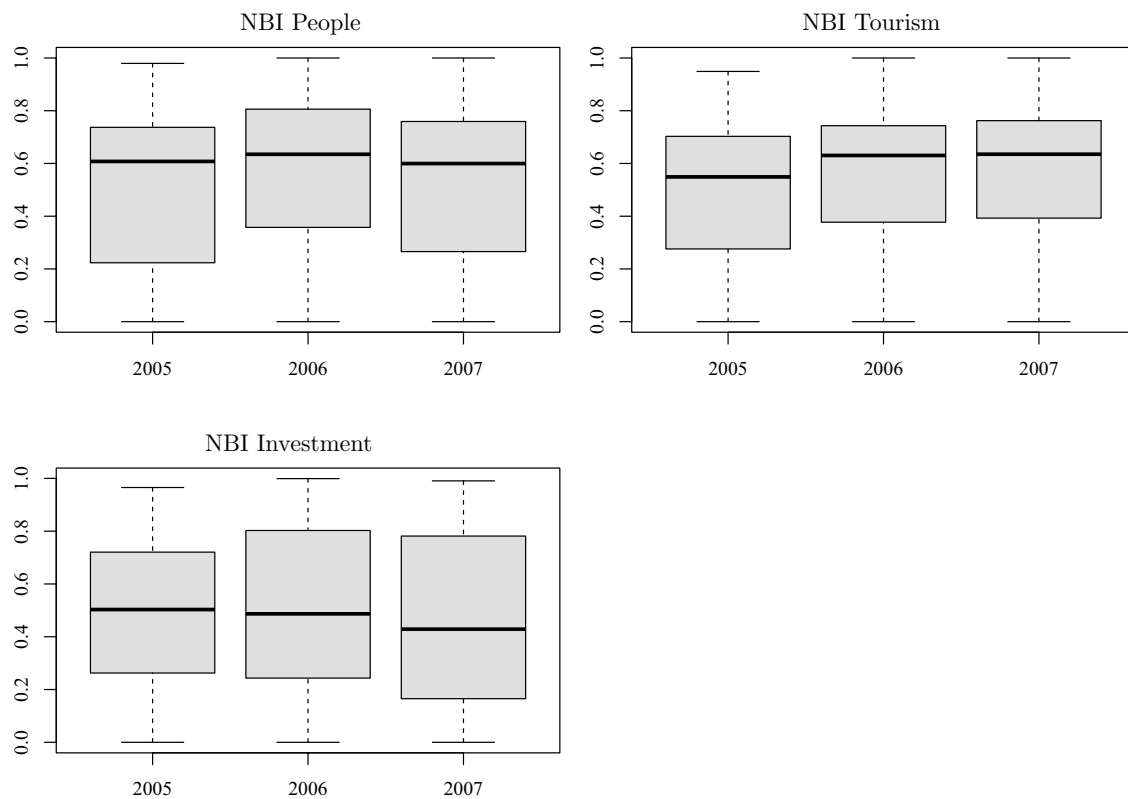


Fig. 4 (continued)

Country list

See Table 1.

Full model results

See Figs. 5, 6 and 7.

Table 1 List of Countries included in sample $N = 40$

Argentina	Australia	Belgium	Brazil	Canada
China	Czech Republic	Denmark	Egypt	Estonia
Finland	France	Germany	Greece	Hungary
India	Indonesia	Iran	Ireland	Israel
Italy	Japan	South Korea	Latvia	Lithuania
Malaysia	Mexico	Netherlands	New Zealand	Norway
Poland	Portugal	Russian Federatio	Singapore	Spain
Sweden	Switzerland	Turkey	United Kingdom	USA



Fig. 5 Estimates of network control terms in Poisson ERGMs. Bars span 95% confidence intervals. Some confidence intervals are not visible due to being small and the large range

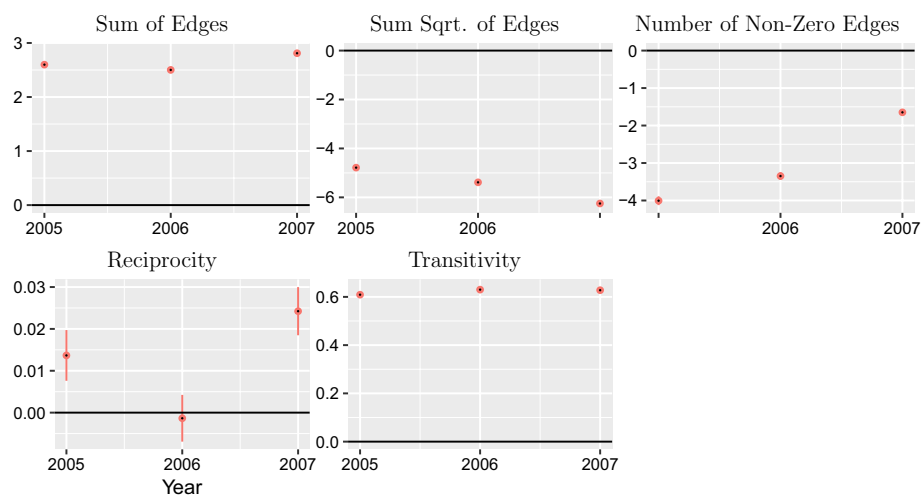


Fig. 6 Estimates of exogenous edge terms in Poisson ERGMs. Bars span 95% confidence intervals. Some confidence intervals are not visible due to being small and the large range

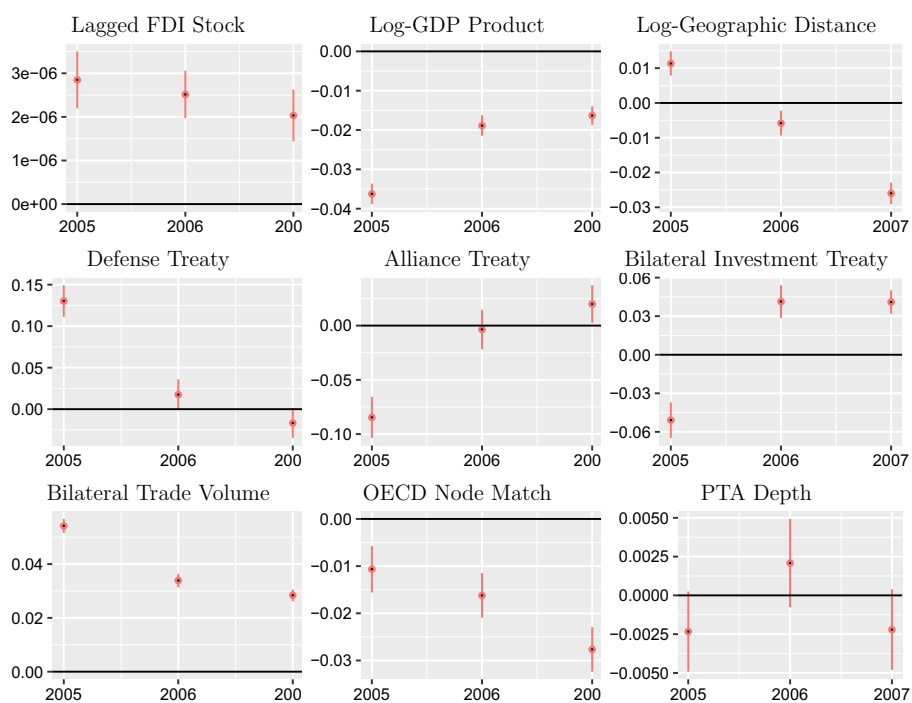
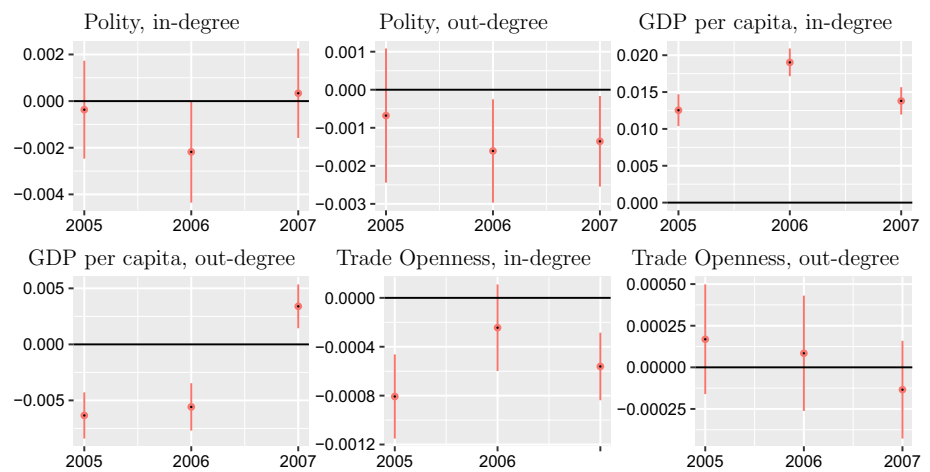


Fig. 7 Estimates of exogenous node terms in Poisson ERGMs. Bars span 95% confidence intervals. Some confidence intervals are not visible due to being small and the large range



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